

PDF Parsing Strategies & Tabular Data Extraction

Lesson 03 · Studio: Grounding Agents in Knowledge · Lyzr University

What you'll learn

- Why layout-aware PDF parsers matter for complex documents
- Using LLM Sherpa to parse tables and visual hierarchies
- How text chunking converts document sections into vector embeddings
- Testing retrieval accuracy in the Knowledge Base Playground before attaching to an agent

Key Concepts

Layout-Aware Parsing

LLM Sherpa preserves headings, table columns, and cell relationships in PDF documents.

Vector Chunking

Extracted text is split into chunks, converted to high-dimensional embeddings, and stored in the vector database.

KB Playground Testing

Run trial queries in the KB Playground to verify similarity scores (%) and chunk quality before deploying.

Code & Workflow Reference

```
# Selecting LLM Sherpa for tabular layout awareness
kb.add_pdf("nova_bridge_handbook.pdf", data_parser="llmsherpa")

# Retrieve & test in playground
results = kb.test_retrieval(query="How many hours of work is expected of a part-time employee?")
for chunk in results:
    print(f"Similarity Match: {chunk.score * 100:.1f}%")
    print(f"Content: {chunk.text}")
```